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C1 - Portfolio Deliverable #4: Exploratory Data Analysis

September 2026

Introduction

The following exploratory data analysis examines how AI occupational exposure (AIOE), projected employment growth, wages, occupational categories, and education requirements interact across U.S. occupations. Using BLS Employment Projections data alongside AIOE measures, the analysis identifies patterns that can help distinguish between occupations that are simply exposed to AI-driven change and those that may face greater economic vulnerability. The visualizations explore these relationships to better understand which factors may contribute to career resilience in an increasingly automated workforce.

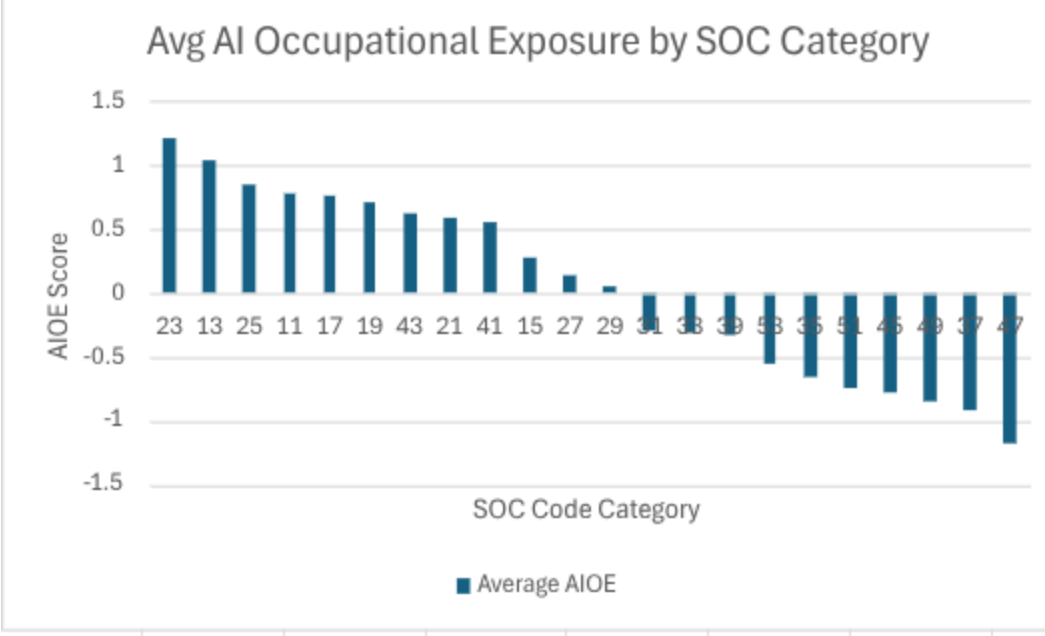
The research questions for the exploratory analysis are as follows:

Question 1: Which U.S. occupations face the highest automation risk?

Question 2: What's the relationship between earnings, projected job growth, & AIOE?

Question 3: How can individuals identify economically resilient versus high-risk careers?

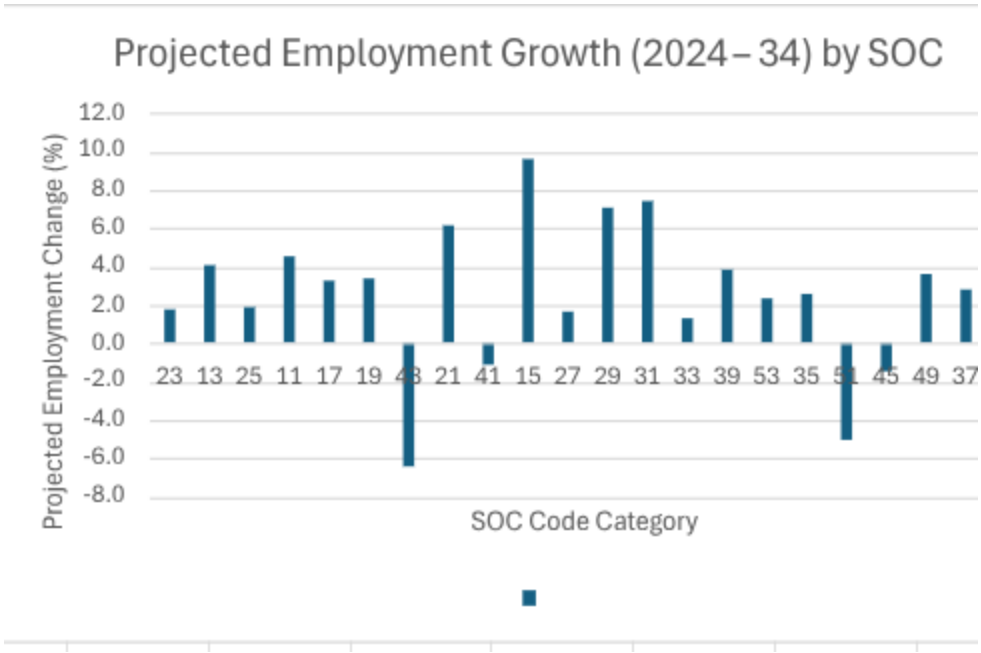
Exploratory Data Analysis:



Visualization #1 Analysis : AIOE by SOC → Which groups have low AI/OE exposure?

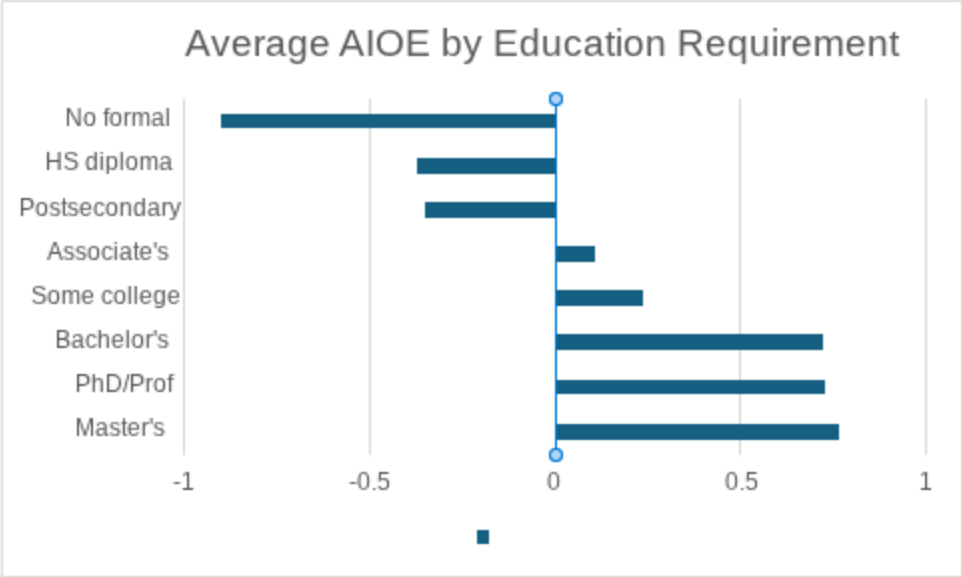
Average AIOE scores vary substantially across SOC major occupational groups. Legal occupations (SOC 23) and Business and Financial Operations occupations (SOC 13) have the highest average AIOE scores, while several occupational groups characterized by more physical or manual work appear toward the lower end of the distribution. Overall, the visualization suggests greater AI exposure among occupations involving information processing, language, analysis, and other cognitive tasks, while occupations requiring more physical, hands-on, or environment-dependent activities tend to demonstrate lower exposure.

* Represents AI occupational exposure rather than likelihood an occupation will be eliminated.



Visualization 2 Analysis: Projected % growth by Occupational groups → How do outlooks differ?

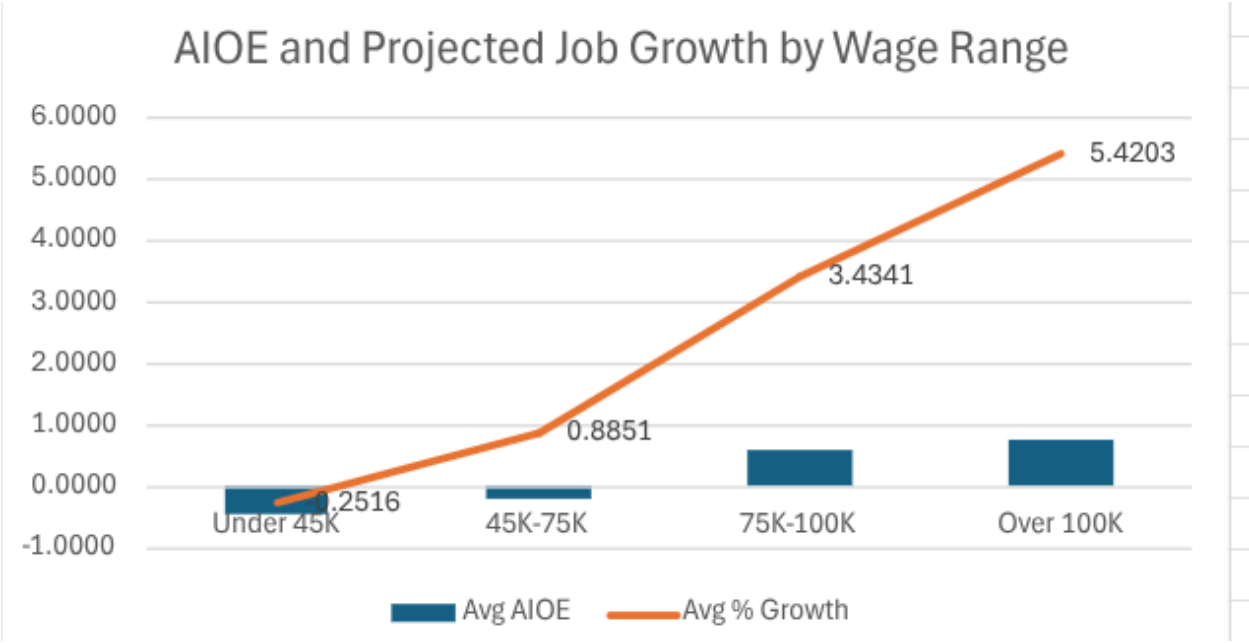
Projected employment growth varies considerably across SOC occupational groups. Computer and Mathematical Occupations (SOC 15- such as data scientists, operations research analysts, and computer systems analysts) show the highest average projected growth among the categories shown at approximately 9.6%, showing the computer and mathematical category is projected to grow strongly even though AI is increasingly relevant to many of the tasks performed in those occupations. Legal Occupations (SOC 23) have the highest average AIOE score (~1.21) but are still projected to grow by approximately 1.7%.



Visualization 3 Analysis: Average AIOE By education requirement

Average AI occupational exposure shows a noticeable relationship with the typical education required for entry. Occupations requiring a master's degree have the highest average AIOE score (~0.77), followed closely by doctoral/professional degree (~0.73) and bachelor's degree occupations (~0.72). In comparison, occupations requiring some college but no degree (~0.24) or an associate's degree (~0.11) show substantially lower positive exposure. Overall, the pattern suggests that occupations requiring higher levels of formal education tend to possess greater overlap with current AI capabilities.

These results challenge the assumption that workers in occupations requiring less education necessarily face the greatest AI exposure. In this dataset, the strongest AI overlap appears among occupations requiring bachelor's and advanced degrees. However, a higher AIOE score represents greater exposure or task overlap with AI capabilities, not a direct probability of job displacement. Therefore, projected employment growth needs to be examined before determining occupational groups' employment risk.



Visualization 4 Analysis: How does job growth by wage category relate to AIOE?

There appear to be two simultaneous upward trends as wage level increases. Average AIOE increases substantially, moving from negative values in the two lower-wage groups to positive values in the two higher-wage groups. At the same time, projected employment growth rises from slightly negative for occupations under $45K to 5.42% for occupations paying over $100K. These trends complicate the idea that greater AI exposure automatically means greater job displacement. Based on the data sets analyzed, the higher-wage groups have both greater average AI exposure and stronger projected employment growth.

Conclusion

The analysis indicates that AI exposure varies substantially across occupations and is especially pronounced among occupations requiring higher levels of education. Average AIOE was highest among occupations requiring a Master's degree (0.769), followed by PhD/professional degrees (0.732) and Bachelor's degrees (0.724). In comparison, occupations

requiring no formal education had an average AIOE of −0.895, suggesting substantially lower exposure to AI.

However, high AI exposure should not automatically be interpreted as high risk of job elimination. The employment-growth analysis illustrates this distinction. Computer and Mathematical occupations (SOC 15), for example, have projected employment growth of approximately 9.6% until 2034, the highest among the SOC categories analyzed, despite knowledge-based professional work of this occupational group being relatively exposed to AI. Therefore, AIOE appears to measure the potential for AI to affect occupational tasks more effectively than it predicts whether an occupation will disappear.

The analysis suggests that the relationship between earnings, job growth, and AI exposure is complex rather than directly inverse. Overall, AIOE and projected employment growth had a weak negative correlation of approximately −0.25, meaning greater AI exposure was only weakly associated with lower projected growth across individual occupations.

The wage analysis revealed an additional pattern. Occupations earning under $45,000 had an average AIOE of −0.46 and average projected growth of approximately −0.25%, while occupations earning over $100,000 had the highest average AIOE (0.75) but also the highest average projected growth (5.42%).

Specific occupational categories demonstrate why this distinction matters. Computer and Mathematical occupations (SOC 15) are projected to grow 9.6%, while Healthcare Practitioners and Technical occupations (SOC 29) and Healthcare Support occupations (SOC 31) are projected to grow 7.1% and 7.4%, respectively. In contrast, Office and Administrative Support occupations (SOC 43) are projected to decline 6.4%, while Production occupations (SOC 51) are projected to decline approximately 5.0%. These results suggest that higher earnings and

greater AI exposure can coexist with strong employment growth. AI exposure alone therefore does not determine an occupation's economic outlook.

The findings suggest that career resilience should be evaluated using multiple indicators together: AIOE, projected employment growth, wages, education requirements, and occupational category. For example, Computer and Mathematical occupations (SOC 15) show very strong projected growth (9.6%), while the education analysis shows that Bachelor's- and graduate-degree occupations generally have relatively high AI exposure. This represents a career area that may be highly exposed to AI-driven changes without necessarily being economically vulnerable. By comparison, Office and Administrative Support (SOC 43) has projected employment decline of −6.4%, while Production occupations (SOC 51) are projected to decline approximately −5.0%. Occupations that combine declining employment projections with substantial automation exposure would warrant greater attention than occupations that simply have a higher AIOE.

The data ultimately suggests that individuals can and should distinguish between exposure and vulnerability. High AIOE may indicate that the *tasks within a job* are likely to change, while the combination of high exposure, weak occupational growth projection , and weaker economic outcomes provides a stronger indication of potential career vulnerability.

Overall, this exploratory analysis suggests that AI exposure does not automatically translate into job displacement. AIOE was considerably higher among occupations requiring Bachelor's and graduate degrees, demonstrating that AI exposure extends strongly into highly educated, knowledge-based work. At the same time, several occupational categories showed strong employment projections, including Computer and Mathematical (9.6%), Healthcare Support (7.4%), Healthcare Practitioners and Technical (7.1%), and Community and Social Service occupations (6.2%). Conversely, Office and Administrative Support (−6.4%) and

Production occupations (approximately −5.0%) showed substantial projected declines. The wage analysis further demonstrated that occupations earning over $100,000 had both the highest average AIOE (0.75) and the strongest average projected growth (5.42%). Combined with the weak overall correlation between AIOE and projected growth ($r \approx -0.25$), these findings suggest that automation vulnerability is multidimensional. Evaluating AI exposure alongside wages, education requirements, occupational category, and projected employment growth provides a more meaningful assessment of career resilience than automation exposure alone.

References

File Name: occupation.xlsx

Description:U.S Bureau of Labor Statistics 2024

Dataset Details: 1119 Rows & 16 Columns

Size: 398 KB

Source: (U.S) Bureau of Labor Statistics - Occupational projections, 2024–2034, and worker characteristics, 2024 (Employment in thousands)

AIOE Dataset *

FIle Name: AIOE_DataAppendix.xlsx

Description: AIOE Data- Appendix A

Dataset details: 775 rows & 3 columns

Size: 167 KB

Source: GitHub- AIOE Data

* AIOE stands for Artificial Intelligence Occupational Exposure.